



LaSalle County Station

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10 CFR 50.73

RA17-017

March 17, 2017

U.S. Nuclear Regulatory Commission  
ATTN: Document Control Desk  
Washington, DC 20555-0001

LaSalle County Station, Units 1 and 2  
Renewed Facility Operating License Nos. NPF-11 and NPF-18  
NRC Docket Nos. 50-373 and 50-374

Subject: Licensee Event Report 2017-002-00, Secondary Containment Inoperable  
Due to Interlock Doors Open

In accordance with 10 CFR 50.73(a)(2)(v)(C) and 10 CFR 50.73(a)(2)(v)(D), Exelon Generation Company, LLC (EGC) is submitting Licensee Event Report (LER) Number 2017-002-00 for LaSalle County Station, Units 1 and 2.

There are no regulatory commitments in this letter. Should you have any questions concerning this report, please contact Mr. Guy V. Ford, Jr., Regulatory Assurance Manager, at (815) 415-2800.

Respectfully,

A handwritten signature in black ink that reads "Harold Vinyard".

Harold T. Vinyard  
Plant Manager  
LaSalle County Station

Enclosure: Licensee Event Report

cc: Regional Administrator – NRC Region III  
NRC Senior Resident Inspector – LaSalle County Station

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<http://www.nrc.gov/reading-rm/doc-collections/nureqs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to [Infocollections.Resource@nrc.gov](mailto:Infocollections.Resource@nrc.gov), and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

**1. FACILITY NAME**

LaSalle County Station, Unit 1

**2. DOCKET NUMBER**

05000373

**3. PAGE**

1 OF 4

**4. TITLE**

Secondary Containment Inoperable Due to Interlock Doors Open

| 5. EVENT DATE |     |      | 6. LER NUMBER |                   |         | 7. REPORT DATE |     |      | 8. OTHER FACILITIES INVOLVED   |               |
|---------------|-----|------|---------------|-------------------|---------|----------------|-----|------|--------------------------------|---------------|
| MONTH         | DAY | YEAR | YEAR          | SEQUENTIAL NUMBER | REV NO. | MONTH          | DAY | YEAR | FACILITY NAME                  | DOCKET NUMBER |
| 01            | 18  | 2017 | 2017          | - 002             | - 00    | 03             | 17  | 2017 | LaSalle County Station, Unit 2 | 05000374      |
|               |     |      |               |                   |         |                |     |      | FACILITY NAME                  | DOCKET NUMBER |
|               |     |      |               |                   |         |                |     |      | NA                             | NA            |

| 9. OPERATING MODE | 11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply) |                                             |                                                       |                                               |
|-------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------|-----------------------------------------------|
| 1                 | <input type="checkbox"/> 20.2201(b)                                                           | <input type="checkbox"/> 20.2203(a)(3)(i)   | <input type="checkbox"/> 50.73(a)(2)(ii)(A)           | <input type="checkbox"/> 50.73(a)(2)(viii)(A) |
|                   | <input type="checkbox"/> 20.2201(d)                                                           | <input type="checkbox"/> 20.2203(a)(3)(ii)  | <input type="checkbox"/> 50.73(a)(2)(ii)(B)           | <input type="checkbox"/> 50.73(a)(2)(viii)(B) |
|                   | <input type="checkbox"/> 20.2203(a)(1)                                                        | <input type="checkbox"/> 20.2203(a)(4)      | <input type="checkbox"/> 50.73(a)(2)(iii)             | <input type="checkbox"/> 50.73(a)(2)(ix)(A)   |
|                   | <input type="checkbox"/> 20.2203(a)(2)(i)                                                     | <input type="checkbox"/> 50.36(c)(1)(i)(A)  | <input type="checkbox"/> 50.73(a)(2)(iv)(A)           | <input type="checkbox"/> 50.73(a)(2)(x)       |
| 10. POWER LEVEL   | <input type="checkbox"/> 20.2203(a)(2)(ii)                                                    | <input type="checkbox"/> 50.36(c)(1)(ii)(A) | <input type="checkbox"/> 50.73(a)(2)(v)(A)            | <input type="checkbox"/> 73.71(a)(4)          |
|                   | <input type="checkbox"/> 20.2203(a)(2)(iii)                                                   | <input type="checkbox"/> 50.36(c)(2)        | <input type="checkbox"/> 50.73(a)(2)(v)(B)            | <input type="checkbox"/> 73.71(a)(5)          |
|                   | <input type="checkbox"/> 20.2203(a)(2)(iv)                                                    | <input type="checkbox"/> 50.46(a)(3)(ii)    | <input checked="" type="checkbox"/> 50.73(a)(2)(v)(C) | <input type="checkbox"/> 73.77(a)(1)          |
|                   | <input type="checkbox"/> 20.2203(a)(2)(v)                                                     | <input type="checkbox"/> 50.73(a)(2)(i)(A)  | <input checked="" type="checkbox"/> 50.73(a)(2)(v)(D) | <input type="checkbox"/> 73.77(a)(2)(i)       |
|                   | <input type="checkbox"/> 20.2203(a)(2)(vi)                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(B)  | <input type="checkbox"/> 50.73(a)(2)(vii)             | <input type="checkbox"/> 73.77(a)(2)(ii)      |
|                   | <input type="checkbox"/> 50.73(a)(2)(i)(C)                                                    | <input type="checkbox"/> OTHER              | Specify in Abstract below or in NRC Form 366A         |                                               |

**12. LICENSEE CONTACT FOR THIS LER**

## LICENSEE CONTACT

Joe Fiesel, Maintenance Director

## TELEPHONE NUMBER (Include Area Code)

(815) 415-2500

**13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT**

| CAUSE | SYSTEM | COMPONENT | MANU-FACTURER | REPORTABLE TO EPIX | CAUSE | SYSTEM | COMPONENT | MANU-FACTURER | REPORTABLE TO EPIX |
|-------|--------|-----------|---------------|--------------------|-------|--------|-----------|---------------|--------------------|
| B     | NG     | IMEC      | SDC           | Y                  | NA    | NA     | NA        | NA            | NA                 |

**14. SUPPLEMENTAL REPORT EXPECTED**☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO**15. EXPECTED SUBMISSION DATE**

| MONTH | DAY | YEAR |
|-------|-----|------|
| NA    | NA  | NA   |

**ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)**

On January 18, 2017, LaSalle County Station, Unit 1 was in Mode 1 at 100 percent power, and Unit 2 was in Mode 1 at 100 percent power with movement of irradiated fuel (MIF) in progress. At 2056 hours CST, both air-lock doors of the Unit 2 reactor building 710-foot elevation three-way air-lock opened simultaneously for approximately five seconds during personnel ingress. Both doors were immediately secured in the interlock, and MIF activities were immediately suspended. While both interlock doors were open, Technical Specifications (TS) Surveillance Requirement (SR) 3.6.4.1.2 to verify one secondary containment access door in each access opening is closed was not met. Secondary containment was declared inoperable for the period of time that both interlock doors were open. TS 3.6.4.1 Required Action (RA) A.1 to restore secondary containment to OPERABLE status within four hours was entered and exited. TS 3.6.4.1 RA C.1 to immediately suspend MIF in secondary containment was entered and exited.

The most probable cause of the interlock failure was the intermittent failure of the relays on the door controller circuit card based on previous history with these circuit boards. Previous events and prior causal investigations have indicated degradations in the relays for these circuit cards, and the corrective actions to replace interlock door circuit cards was ongoing at the time this event occurred. The controller circuit card was replaced, which restored the interlock functionality on January 19, 2017.

**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to [Infocollections.Resource@nrc.gov](mailto:Infocollections.Resource@nrc.gov), and to the Desk Officer, Office of Information and Regulatory Affairs, NEOF-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

| 1. FACILITY NAME               | 2. DOCKET NUMBER | 3. LER NUMBER |                   |         |
|--------------------------------|------------------|---------------|-------------------|---------|
|                                |                  | YEAR          | SEQUENTIAL NUMBER | REV NO. |
| LaSalle County Station, Unit 1 | 05000373         | 2017          | - 002             | - 00    |

**NARRATIVE****PLANT AND SYSTEM IDENTIFICATION**

LaSalle County Station Unit 1 and Unit 2 are each a General Electric Boiling Water Reactor with 3546 Megawatts Thermal Rated Core Power. The secondary containment is the reactor building, common to both reactor units, which contains the reactor auxiliary and serving equipment for the two reactors. The secondary containment access doors, or air-locks, provide a means for personnel and equipment to enter or exit the reactor building without breaching secondary containment integrity.

**CONDITION PRIOR TO EVENT**

|                        |                                               |                                      |
|------------------------|-----------------------------------------------|--------------------------------------|
| Unit(s): 1 / 2         | Event Date: January 18, 2017                  | Event Time: 2056 CST                 |
| Reactor Mode(s): 1 / 1 | Mode(s) Name: Power Operation/Power Operation | Power Level: 100 percent/100 percent |

**DESCRIPTION OF EVENT**

On January 18, 2017, LaSalle County Station, Unit 1 was in Mode 1 at 100 percent power, and Unit 2 was in Mode 1 at 100 percent power with movement of irradiated fuel (MIF) in progress. At 2056 hours CST, both air-lock doors of the Unit 2 reactor building to the diesel generator corridor (doors 507 and 402) were opened simultaneously for approximately five seconds during personnel ingress. The employee immediately secured both doors in the interlock and notified the Main Control Room (MCR). Shift Operations personnel subsequently blocked access until repairs could be made, and MIF activities were immediately suspended.

While both interlock doors were open, Technical Specifications (TS) Surveillance Requirement (SR) 3.6.4.1.2 to verify one secondary containment access door closed was not met. Secondary containment was declared inoperable for the period of time that both interlock doors were open. Reactor building differential pressure, as observed in the MCR, remained less than -0.25 inches vacuum water gauge at all times. TS 3.6.4.1 Required Action (RA) A.1 to restore secondary containment to OPERABLE status within four hours was entered and exited. TS 3.6.4.1 RA C.1 to immediately suspend MIF in secondary containment was entered and exited.

**CAUSE OF EVENT**

The cause of the interlock failure was the intermittent failure of a circuit board which is designed to prevent more than one door to be open at a time. Similar previous interlocked door circuit board failures have been identified, through formal failure analysis, to have been caused by degraded relays integral to the circuit board. A causal investigation for an event in 2015 first identified interlock circuit cards as a cause, with a causal investigation for an event in 2016 identifying that relays on the circuit card as the specific cause of the circuit cards. Corrective actions from the 2016 investigation had been ongoing to upgrade interlock door circuit cards with ones that had improved relays. The interlock that failed was scheduled to have its circuit card replaced the week following this event as a corrective action from the previous interlock failure event in 2016.

**REPORTABILITY AND SAFETY ANALYSIS**

This event is reportable in accordance with 10 CFR 50.73(a)(2)(v)(C) and 10 CFR 50.73(a)(2)(v)(D) as an event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident. These doors are ground elevation personnel air-lock between the common diesel generators' corridor to the Unit 2 reactor building. The secondary containment access doors, or air-locks, provide a means for personnel and equipment to enter or exit the reactor building without breaching secondary containment integrity. This is necessary so that the negative pressure can be maintained.

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**NARRATIVE**

The safety significance of this event was minimal, since the reactor building to outside differential pressure remained negative throughout the period that the secondary containment was inoperable. The secondary containment was inoperable for approximately five seconds, which was significantly less than the four-hour Completion Time to restore secondary containment to operable status allowed by TS 3.6.4.1 Required Action A.1.

The function of the secondary containment is to contain, dilute, and hold up fission products that may leak from the primary containment following a Design Basis Accident (DBA). A technical evaluation determined that the inadvertent simultaneous opening of secondary containment doors for less than 30 seconds are bounded by the existing drawdown analysis and dose calculations and will not result in a failure of a safety system function needed to control the release of radioactive material to the environment. The time that both doors were simultaneously opened for this event was approximately five seconds, which is bounded by the technical evaluation (reference Engineering Change (EC) 396711, "Safety System Functional Analysis for Unit 1 & 2 Secondary Containment Air-lock Doors.")

This event did not result in the reactor enclosure differential pressure dropping below the design bases set point of -0.25 inches vacuum water gauge. Both the inner and outer doors were promptly closed by station personnel, which ended the event. This event did not involve any kind of door or air-lock material condition preventing door closure. Additionally, both the inner and outer doors were closed by normal expected means and were capable of remaining closed as designed.

The computed dose for EC 396711 was based on the door opening during the 780 second time period prior to Standby Gas Treatment (SBGT) system drawdown and filtration. This discounts the initial 120 seconds of an event where no release takes place, in accordance with calculation L-003068, "Re-Analysis of Loss of Coolant Accident (LOCA) Using Alternative Source Terms."

The approximate five second opening of the secondary containment doors is bounded by calculation L-003068. Should an event occur, in which both secondary containment doors were open simultaneously for 30 seconds or less, it would result in a potential dose increase of approximately 3.85 percent. The 3.85 percent decrease in margin is inconsequential in comparison to the 10 CFR 100 regulatory limits.

EC 396711 also evaluated the pressure impact on the secondary containment and the ability of the SBGT system to achieve the TS required negative pressure. The results of the evaluation show SBGT would restore secondary containment pressure within three minutes which is well below the 15 minute maximum drawdown time required by TS.

Based on the short duration of door opening (approximately five seconds), no material condition preventing door closure or maintaining the doors closed, and attendance by knowledgeable personnel who closed the doors immediately, the secondary containment safety function was maintained.

**CORRECTIVE ACTIONS**

The defective circuit board with a degraded relay, which had been in-service since February 2016, was replaced with an interim circuit board that had been manufactured with a newer version of the original relay. This circuit board had been previously obtained from the vendor as a contingency for use in the event of a door failure. Operability of the secondary containment was restored on January 19, 2017, at 1325 CST, upon completion of a satisfactory test of the interlock doors. Station personnel were unable to reproduce the failure on the removed card.

A set of final upgraded circuit boards with more robust relays was manufactured, bench-tested by the manufacturer, and "burned in" for 100 hours to ensure functionality, before they were delivered the site for use as the final corrective action repairs. The site performed additional bench-testing of the delivered cards before installation in the plant. The affected interlock doors, as well as the remaining sets of high-traffic interlock doors, were upgraded with the new-style circuit board.

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**NARRATIVE****PREVIOUS OCCURRENCES**

The interlock that failed was scheduled to have the circuit card replaced the week following this event as an action created from a previous failure. Corrective actions from previous events had been ongoing to upgrade interlock door circuit cards with ones that had improved relays. These corrective actions were the result of causal investigations with previous events as follows.

**LER 2016-001-00:**

On February 17, 2016, Unit 2 was in Mode 1 at 100 percent power and Unit 1 was in Mode 5 for a refueling outage with no fuel movements or operation with the potential to drain the reactor vessel (OPDRV) in progress. At 1035 hours CST, both air-lock doors of the Unit 2 Chemistry Lab corridor to Unit 2 reactor building were open at the same time for approximately five seconds. The cause was failure of the relays on the door controller circuit card. The controller circuit card was replaced, which restored the interlock functionality. Corrective actions included determination of the cause of vendor quality issues with the controller circuit card relays and procurement of a more reliable controller circuit card following cause identification of the relay failures from vendor analysis.

**LER 2015-003-00:**

On February 17, 2015, Unit 1 was in Mode 1 at 100 percent power and Unit 2 was in Mode 5 for a refueling outage with no fuel movements in progress. Activities involving OPDRVs were in progress in the secondary containment on Unit 2. At 1145 hours CST, it was reported that both air-lock doors between the Unit 1 diesel generator corridor and the Unit 1 reactor building were open at the same time for approximately five to ten seconds. The cause was a failure of a controller circuit card in the door interlock logic. The circuit card was replaced, which restored the interlock functionality. Corrective actions included a cause determination for the premature controller circuit card failures and replacing the card with more reliable models.

**LER 2015-001-00:**

On December 12, 2014, both Units 1 and 2 were in Mode 1 at 100 percent power with no fuel movements in progress. At 1324 hours CST, it was reported that both air-lock doors between the Unit 2 diesel generator corridor and the Unit 2 reactor building were open at the same time for approximately ten seconds. The cause was a degradation of the door closure mechanism, and the contributing cause was a less than robust design of the door interlock assembly. Corrective actions from the previous occurrences to identify, procure and install a more robust interlock assembly design were in progress at the time of the event. Additional corrective actions, including periodic preventative maintenance to inspect, tighten, and replace fasteners as necessary, were in place but did not preclude this event. This event did not involve a door controller circuitry reliability issue.

**LER 2014-001-00:**

On February 18, 2014, Unit 1 was in Mode 5 with fuel moves in progress during a refueling outage, and Unit 2 was in Mode 1 at 100 percent power. At 1820 hours CST, it was reported that both air-lock doors between the Unit 2 diesel generator corridor and the Unit 2 reactor building were open at the same time for approximately three seconds. The cause of the event was degradation of the door closure mechanism and door frame seal. A contributing cause was a less than robust design of the door interlock assembly. Corrective actions from the previous occurrences to identify, procure and install a more robust interlock assembly design were still in progress at the time of the event. Additional corrective actions included creating a periodic preventative maintenance task to inspect, tighten, and replace fasteners as necessary. This event did not involve a door controller circuitry reliability issue.

**COMPONENT FAILURE DATA**

Manufacturer: Security Door Controls (SDC)  
Device: UR2-4 Controller Card  
Component ID: 1695558